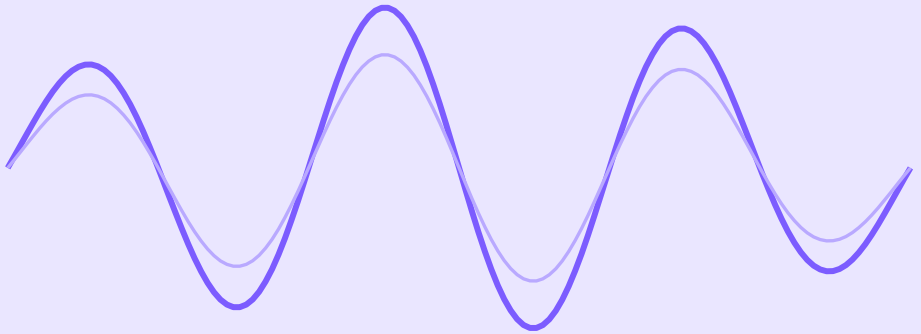
Pulmira

An observation-led respiratory digital twin.

Digital Twin Challenge 2026

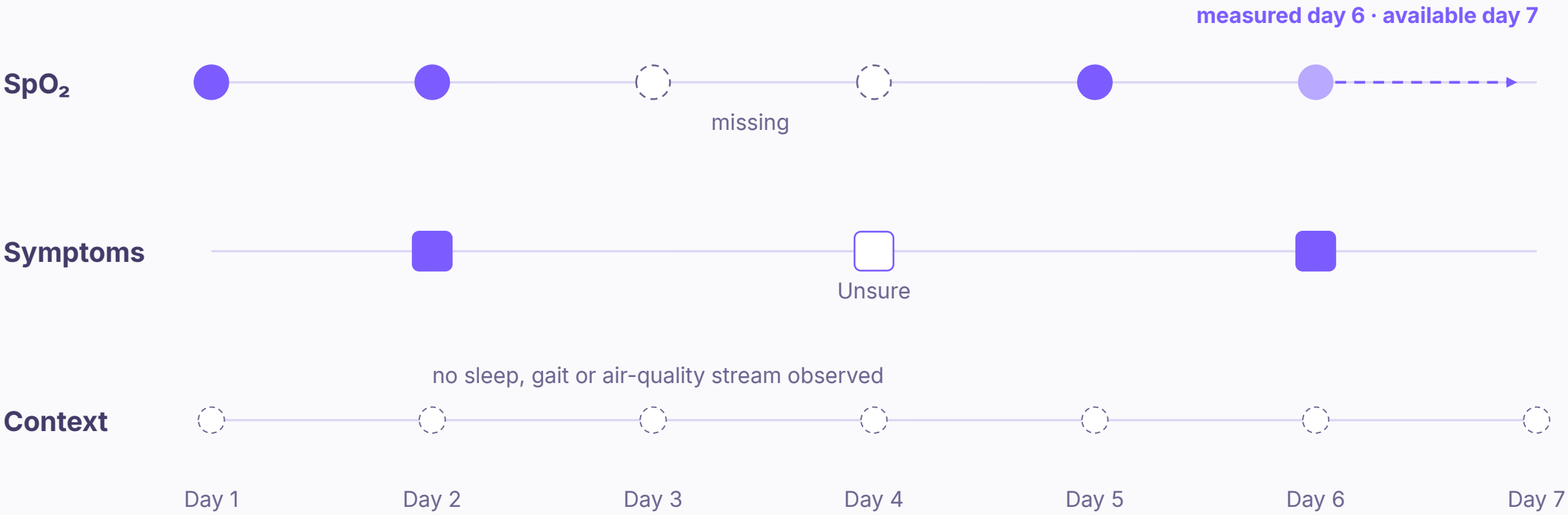
Chennai Institute of Technology



Research prototype · fictional demonstration data

THE PROBLEM

Making respiratory observations useful



An isolated reading says little on its own. Interpretation needs to know which channels exist, how fresh each reading is, and what was still unknown at the decision time.

THE PRODUCT

One connected workspace

01

Monitoring

Fictional profiles, separate device and context streams, coverage review and historical replay.

02

Seven-day formula outlook

A transparent, bounded trend formula. Unvalidated; reported in change points out of 100.

03

Fictional simulator

Sixteen engineering sensitivity scenarios over 180 fictional days.

04

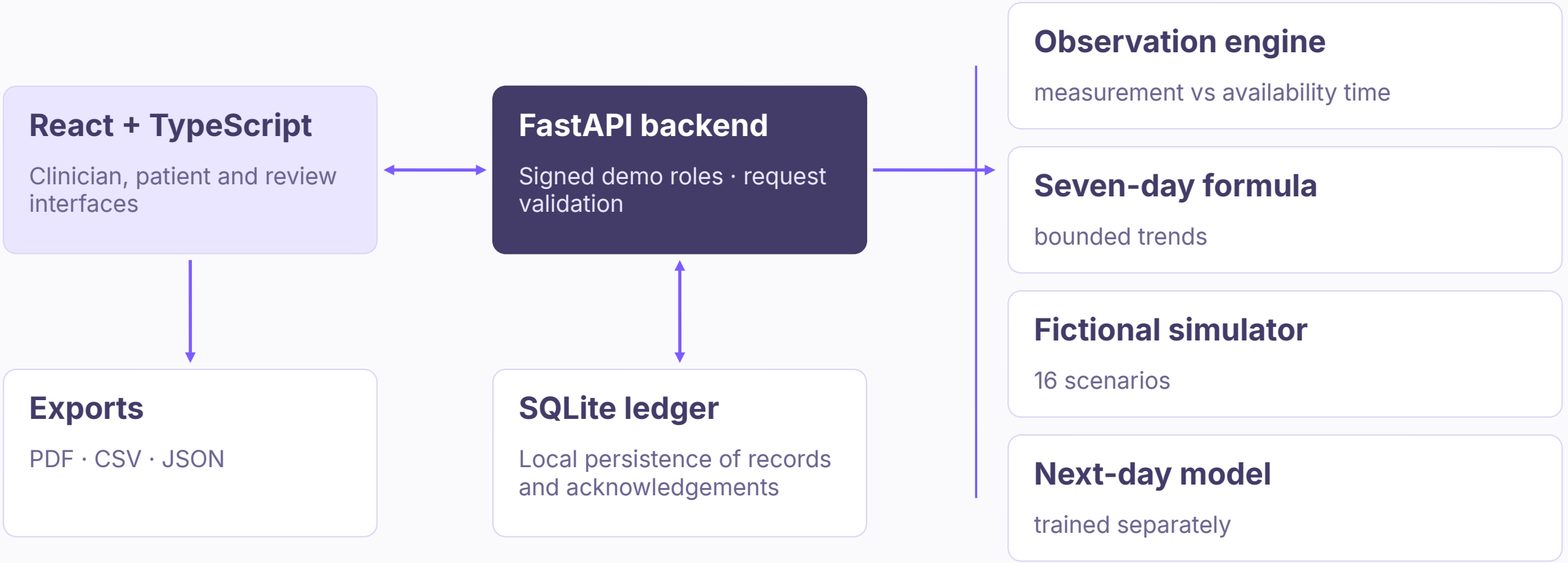
Next-day ML experiment

A separately trained model estimating next-day simulated SpO_2 below 90%.

Shared observation engine and ledger underneath. The three analytical modules never borrow each other's numbers.

ARCHITECTURE

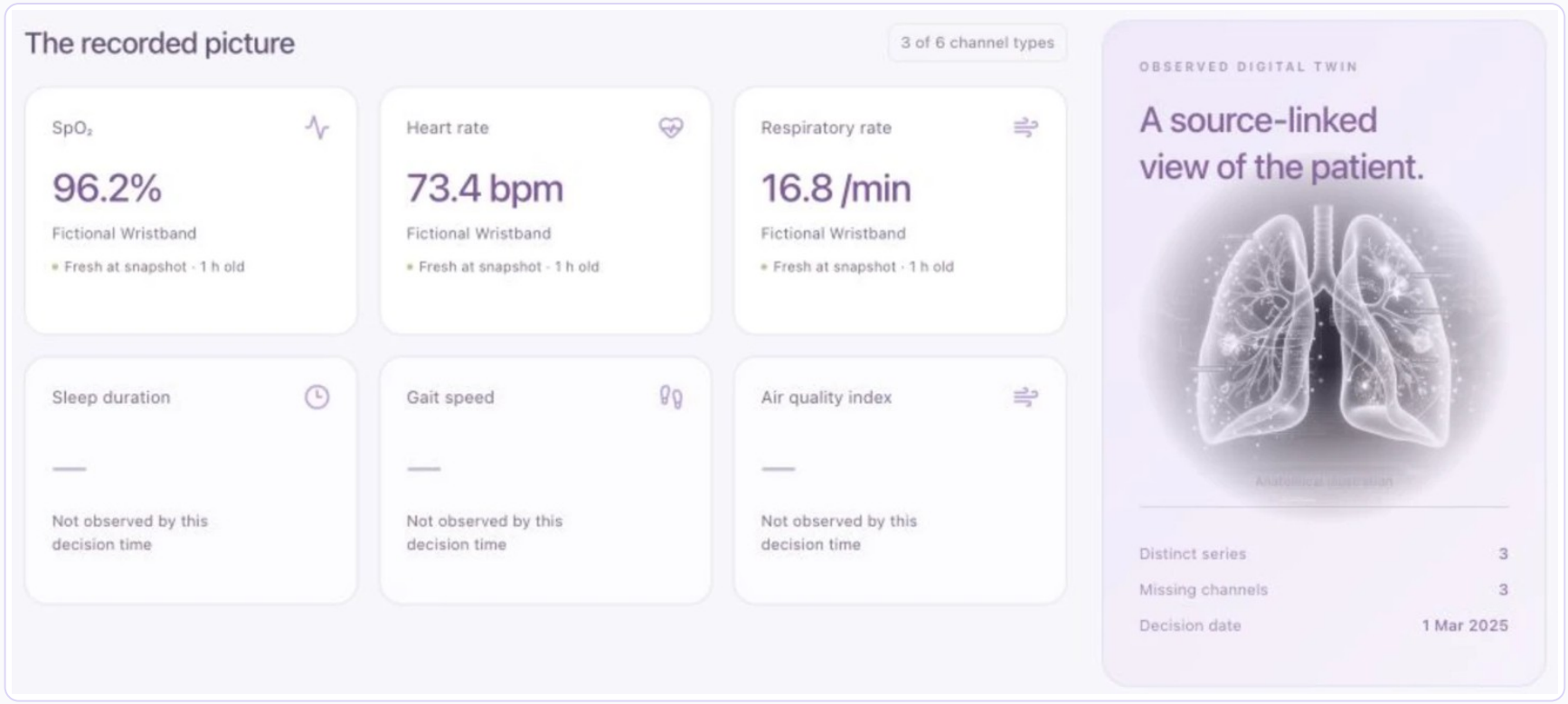
How the layers connect



Results return through the API to the interface. Data enters from historical fictional records; there is no active wearable feed.

PATIENT WORKSPACE

A patient-specific view



Fictional profile at a historical decision time (1 Mar 2025). The lung illustration is decorative anatomy, not a patient scan.

1 Recorded vitals

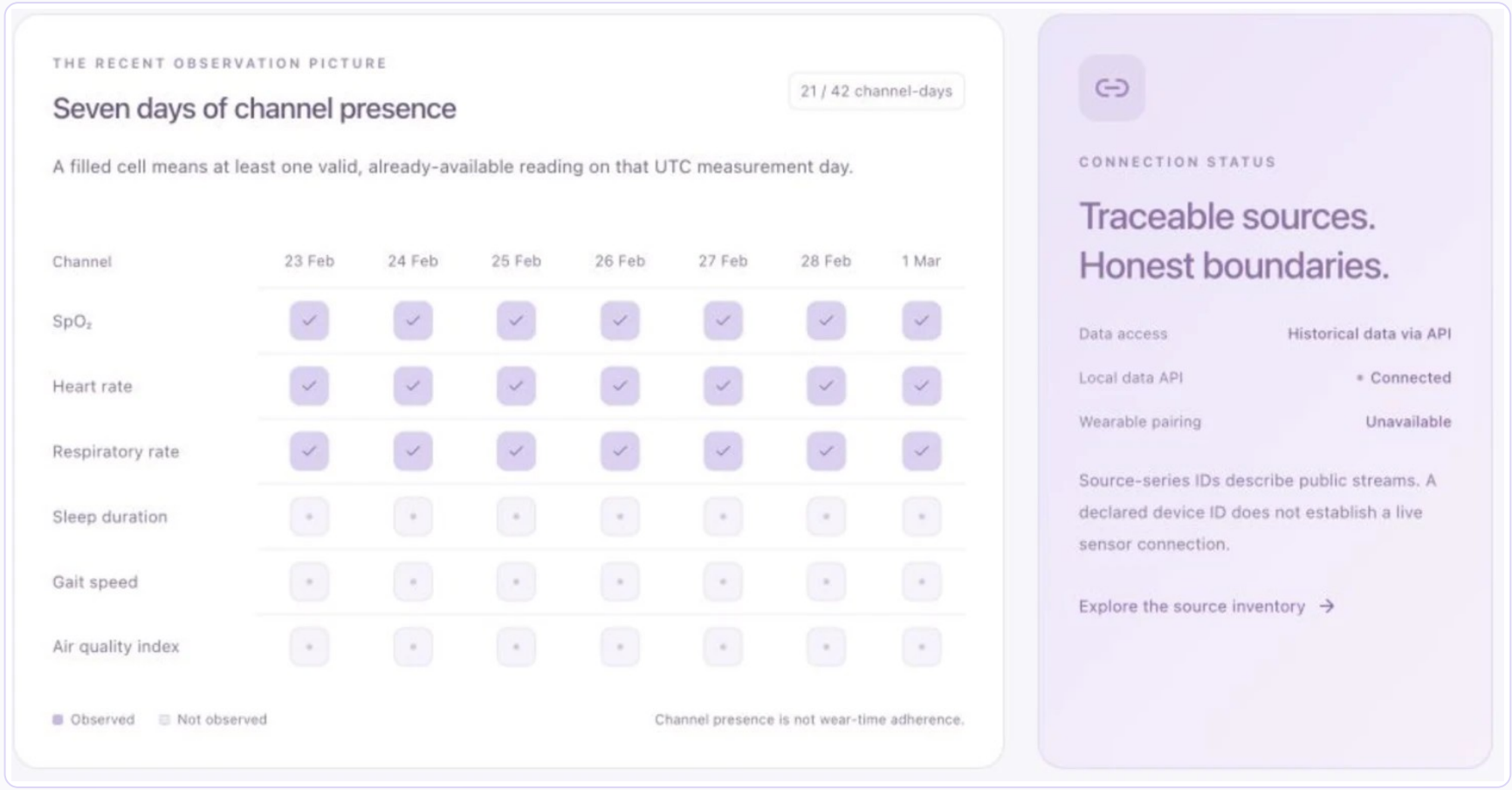
SpO_2 96.2%, heart rate 73.4 bpm and respiratory rate 16.8/min, each with its own source stream and age.

2 Explicit unknowns

Sleep, gait and air quality are not observed by this decision time, so they stay unknown.

DATA QUALITY

Coverage before interpretation



Missing observations remain unknown; they are never treated as normal values. Channel presence is not wear-time adherence.

1 Channel presence

21 of 42 channel-days observed. A filled cell means a valid, already-available reading that UTC day.

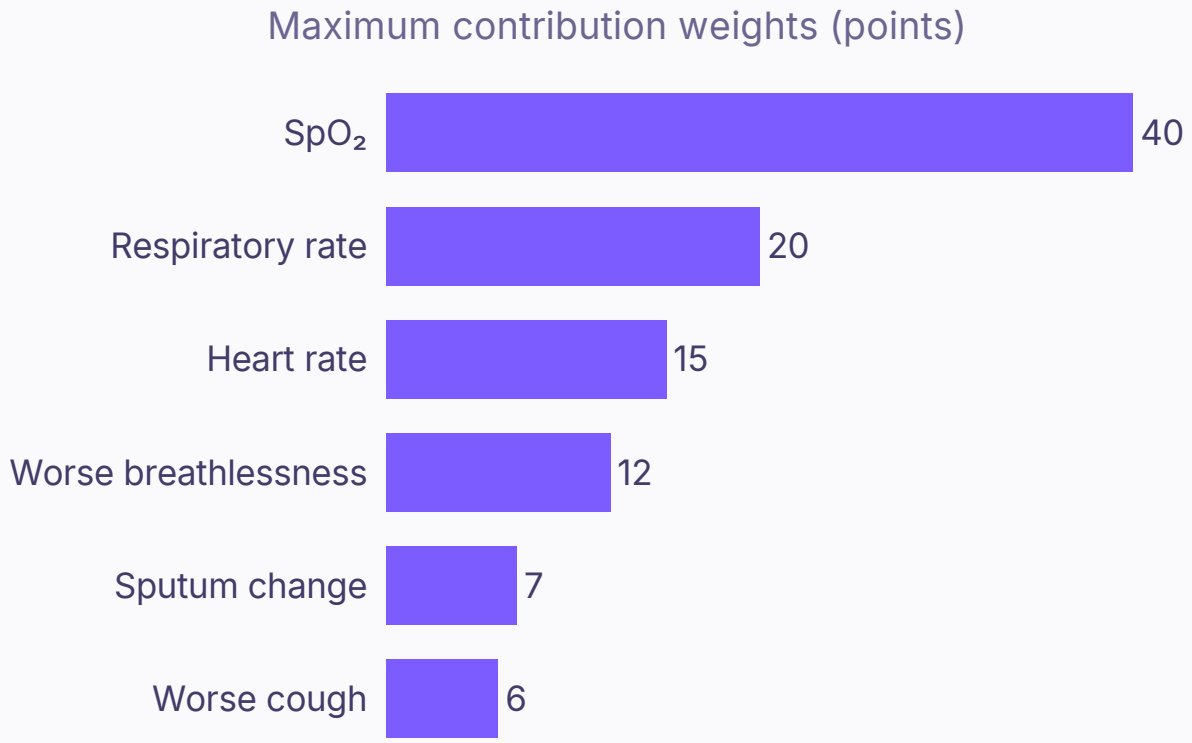
2 API ≠ wearable

The local data API is connected. Physical wearable pairing is unavailable.

SEVEN-DAY OUTLOOK

A transparent seven-day outlook

Unvalidated formula outlook



The historical reference is not a clinically verified stable baseline.

Observation requirements

- ≥4 daily observations spanning 3 days in a 14-day lookback
- Eligible readings within 48 hours
- SpO_2 plus another eligible vital signal

Example peak: 21.8–28.8 change points out of 100. The range comes from unknown inputs, not statistical confidence. Not a probability; predictive accuracy not established.

Seven dates, visible assumptions

THE FULL SEVEN-DAY VIEW

If the recent trend continues.

Partial inputs

Select a day to inspect its projected values and factor contributions.

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
2025-03-02	2025-03-03	2025-03-04	2025-03-05	2025-03-06	2025-03-07	2025-03-08
17.6–24.6	18.3–25.3	19.0–26.0	19.7–26.7	20.4–27.4	21.1–28.1	21.8–28.8
points / 100	points / 100	points / 100	points / 100	points / 100	points / 100	points / 100

SELECTED DAY 7 · 2025-03-08

Projected sensor values

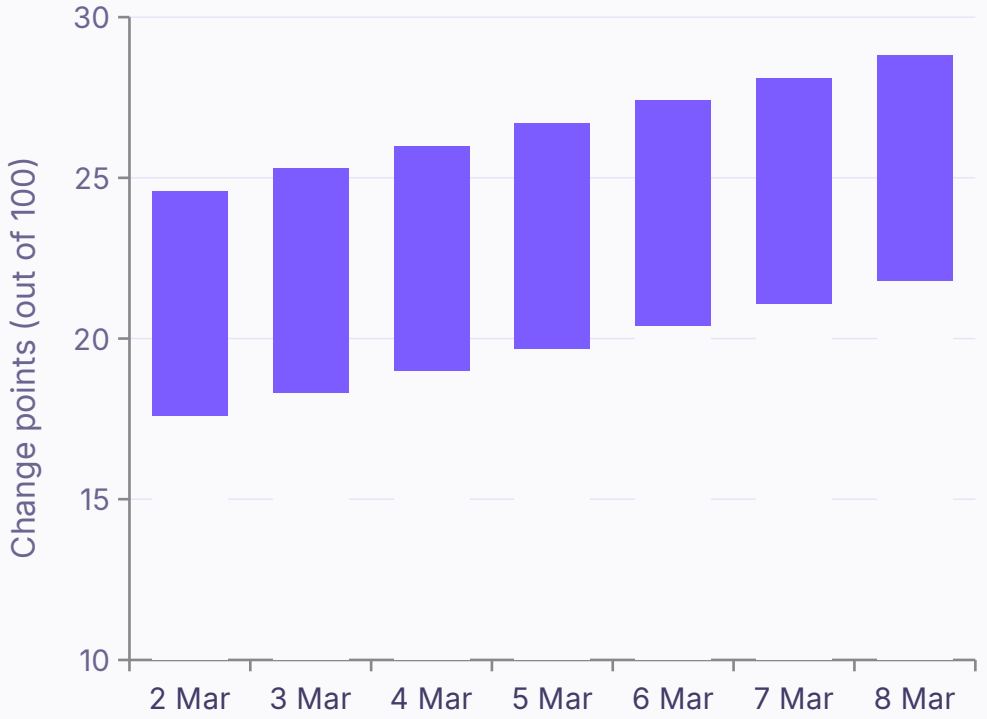
SpO_2	95.95 %
Respiratory rate	17.50 /min
Heart rate	75.50 bpm

Each channel uses its own source stream. No observation from after the decision enters the calculation.

Day 7 formula contributions

Factor	Contribution	Maximum weight
SpO_2	4.2 pts	40 pts
Respiratory rate	3.5 pts	20 pts
Heart rate	2.1 pts	15 pts
Worse breathlessness Recent report held constant	12.0 pts	12 pts

Future values are projections, not recorded observations.



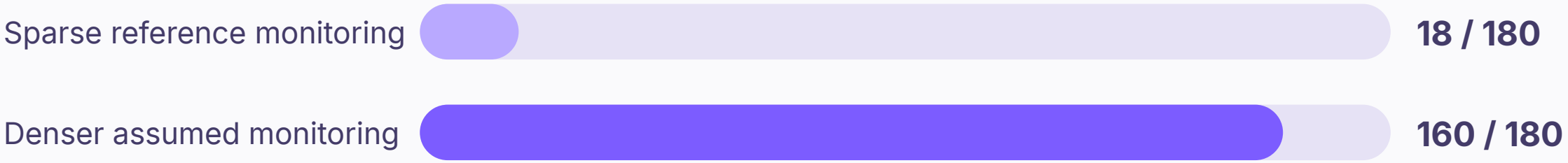
Projected ranges from a 1 Mar 2025, 13:00 UTC decision. Historical fictional example.

The 7-point width reflects the unknown sputum-change answer.

SYNTHETIC SIMULATOR

Testing fictional scenarios

Observed days in a 180-day fictional comparison

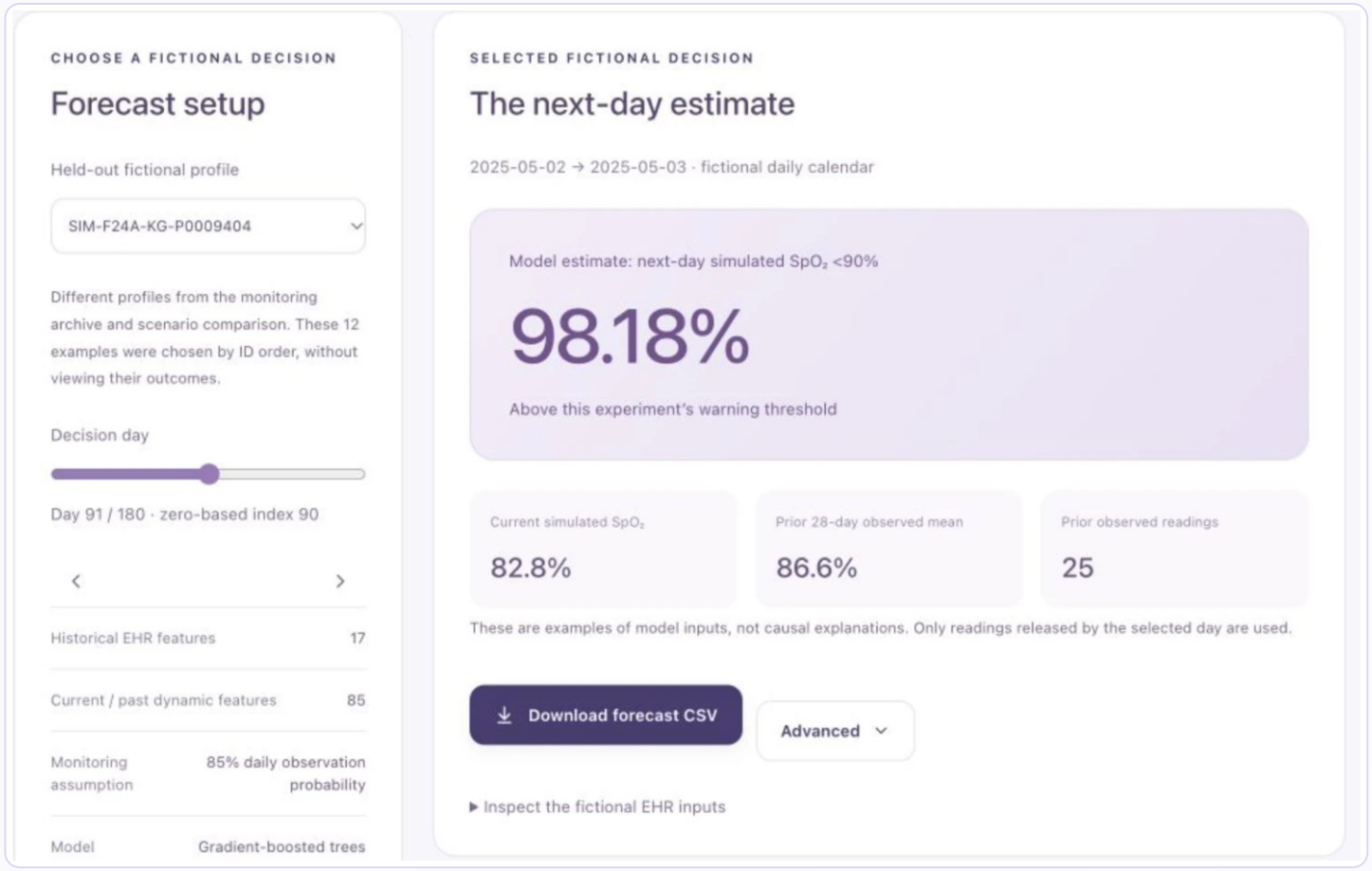


16 engineering sensitivity scenarios vary assumptions such as observation density, then compare what the system can see.

This illustrates an observation-coverage assumption. Changed assumptions do not demonstrate a clinical intervention benefit.

All profiles and values are fictional.

The separate next-day ML model



Fictional held-out profile · clinical validation not established.

102 features

17 static EHR · 85 current/past dynamic

↓

Gradient boosting

160 iterations · depth 3

↓

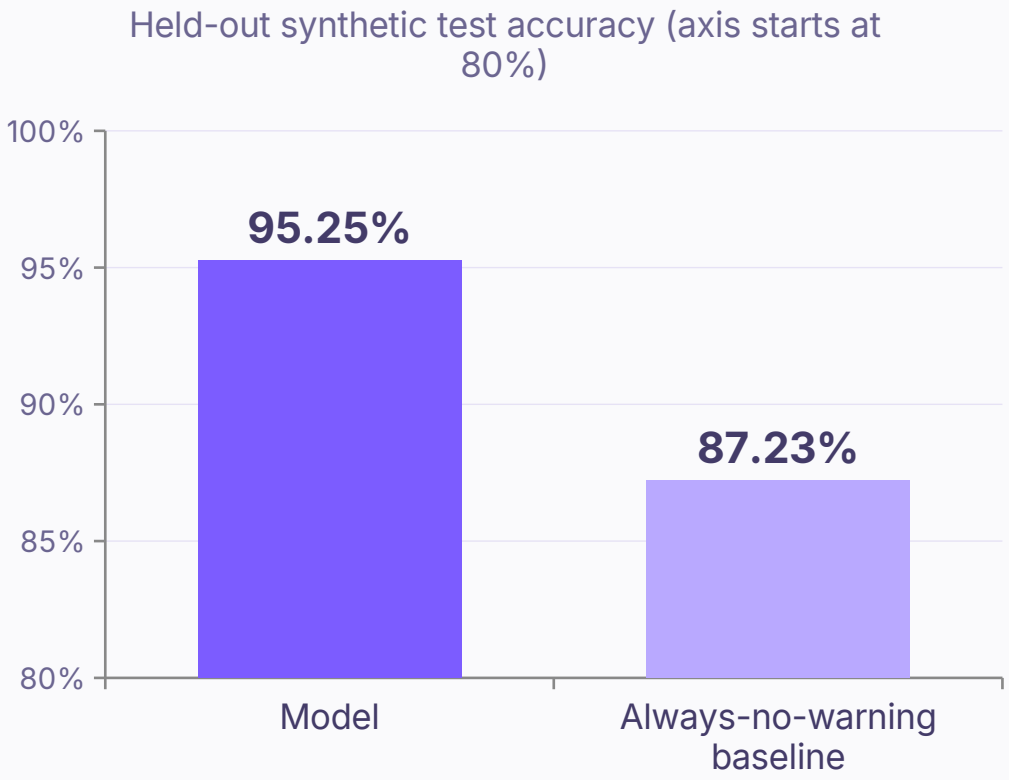
Target

Next-day simulated SpO_2 below 90%

98.18% is one conditional estimate for one fictional profile, not accuracy or confidence. The current reading is already low (82.8%), so this is not early detection. Separate from the seven-day formula.

SYNTHETIC NEXT-DAY SpO_2 EXPERIMENT

What the synthetic evaluation shows



84.38%

Precision

77.11%

Recall

84.70%

Runtime coverage · 24,320 decisions

17,442

Evaluable labelled days · 160 profiles

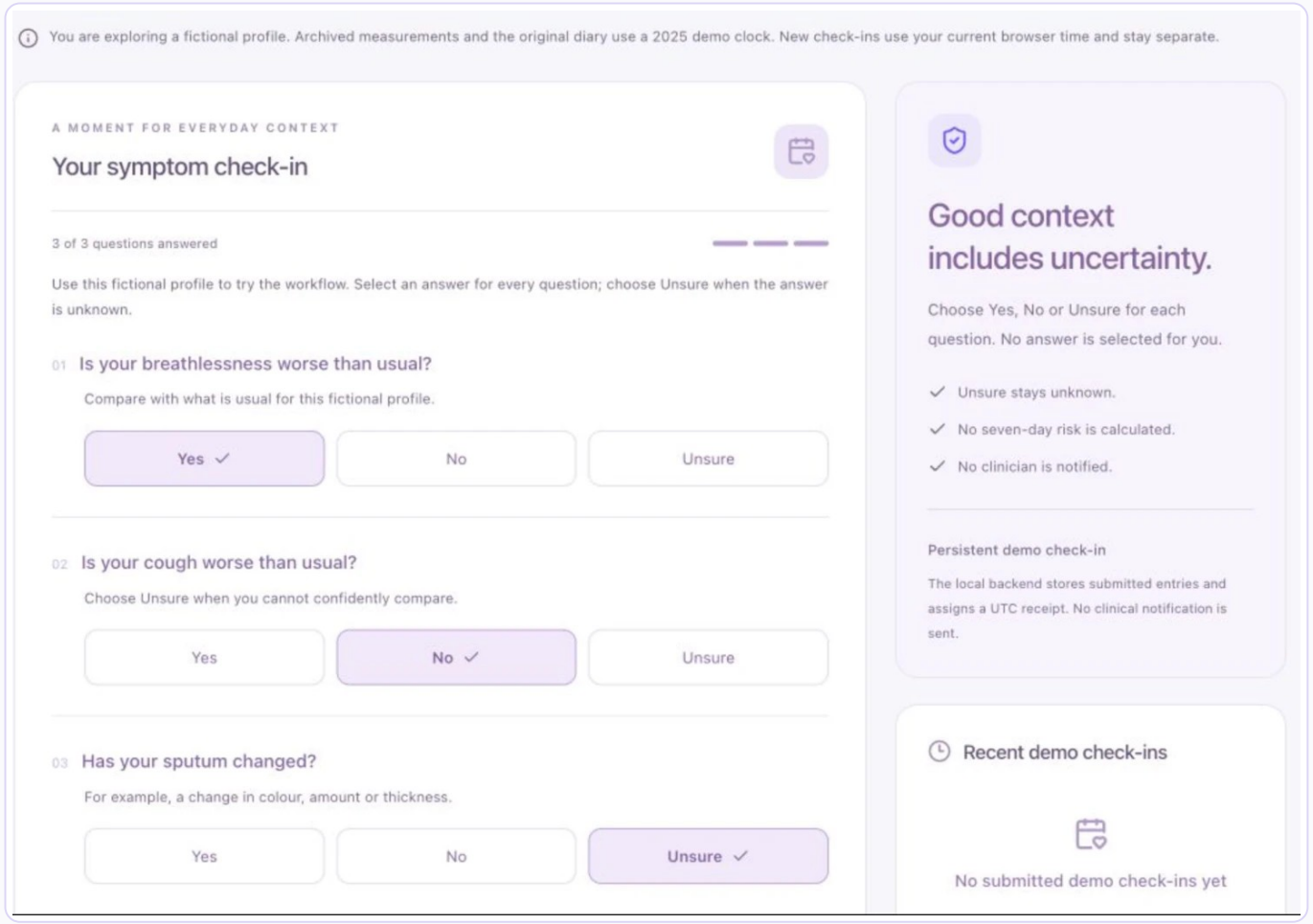
13.48%

Newly low-state recall when the current reading is not already low. Early detection of new deterioration is not established.

Synthetic data only. Not clinical COPD accuracy. Threshold frozen on validation before test.

PATIENT PORTAL

Patient check-in and reporting



Unsubmitted fictional form, shown for illustration.

Yes · No · Unsure

Unsure is stored as unknown, never guessed.

Role-scoped submission

Signed demo roles control who can submit and view.

Separate clocks

Archived data uses a 2025 demo clock; new check-ins use current time.

Exports

PDF, CSV and JSON reports.

EVIDENCE

Evidence and the next validation step

100K-patient synthetic source

Kaggle v3 · Apache-2.0 · hashes verified

→

8,450

fictional COPD profiles

→

8,421

simulator initialisation profiles (29 pre-anchor deaths excluded)

→

800

profiles sampled for the ML experiment

Local archive kept separate, never merged: 86 TOLIFE participants · 53 BIDMC relative-clock recordings. Real-person records are excluded from the public repository.

Next validation step

- Event-linked longitudinal observations with verified timing
- Predeclared, patient-separated evaluation
- Incident-event recall, calibration and independent validation

Software verification

75 automated tests and a production build passed for the public export. This is engineering verification, not clinical validation.

Team code MIT-licensed; upstream dataset terms remain Apache-2.0.



Pulmira

Digital Twin Challenge 2026
Chennai Institute of Technology

TEAM

Guruyugan Karthik

Harishram B

Harihara Srinivas T

REPOSITORY

github.com/GuruyuganKarthik8377/Pulmira

Local research prototype. Fictional demonstration data.